

Software Modeling and UML

Teamwork 3

Activity Diagram Modeling

Objectives

- Understand the role of the activity diagrams
- Be familiar with the general process of modeling activity diagrams
- Become more proficient in using UML tools and master the representation of common modeling elements in activity diagrams

Task 1 Activities in Campus Information Platform (Teamwork)

● Download Courseware

- After students log in, they can download the courseware from the curriculum workspace.
- When a student logs in, the system needs to verify the user's login information. If verified, the system will display the available services. If the authentication fails (with fewer than 3 attempts), the system will return to the login page. If the authentication fails (with 3 or more attempts), the login will be unsuccessful.
- Once the user sees the services provided by the system, they can choose to download the required courseware. After the download is complete, the user exits the system.

● Please draw the activity diagram for the activities mentioned above.

Task 2 Activities in Campus Information Platform (Teamwork)

● Release scores

- After teachers login, they can submit the usual performance of students, they can submit the scores of final exam, they also can submit the scores of practical exercises.
- Once the above scores are submitted, the total scores are calculated and submitted.
- Then, the teacher needs to provide a course summary.
- After review by the academic secretary, the scores will be released.

● Please draw the activity diagram for the activities mentioned above.

Task 3 Activities in Tennis Court Booking System (Teamwork)

●Booking

- Users can select a tennis court online through the app or WeChat and send the booking request.
- The tennis court staff can confirm the booking request.
- Once the information is confirmed, users can make the reservation payment online.
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●Please draw the activity diagram for the activities mentioned above.

Task4 Activities in Hospital Appointment and Registration System

● Making an Appointment

- Patient initiates an appointment request.
- Confirm the available appointment time.
- Patient selects the appointment time and submits the appointment request.
- System confirms the appointment and generates an appointment ID.
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● Please draw the activity diagram for the activities mentioned above.

Steps of Activity Diagram Modeling

- Define the scope of the activity diagram
- Add the initial node and final node
- Add activity nodes
- Add decision nodes
- Identify parallel activities
- Add control flows

Experiment requirements

- The team leader should submit the work through the link provided by the teaching assistant's cloud storage.
- Requirements for the report
 - Report name: UML Teamwork 3 Activity Diagram (Student ID Name, Student ID Name, Student ID Name...) case name
 - Please do not compress any document (不要压缩)
 - The deadline for task submission is 6 PM this Saturday evening
- Make sure to send the report following the specified guidelines; otherwise, we will not be able to receive your submission.